Practical 2

Program –

```
#include <SoftwareSerial.h>

// HC-05: Arduino RX = D10, TX = D11
SoftwareSerial BT(10, 11);

const int ledPin = 2;
const int tempPin = A0;
const float alertTemp = 35.0;

void setup() {
  pinMode(ledPin, OUTPUT);

  Serial.begin(9600);
  BT.begin(9600);

  Serial.println("=== Bluetooth Environment Control ===");
  Serial.println("Commands from mobile app:");
  Serial.println("L = LED ON");
  Serial.println("O = LED OFF");
  Serial.println("T = Read Temperature");
  Serial.println("S = System Status");
  Serial.println("-----------------------------------");
  Serial.println("System Ready!");
}

void loop() {

  if (BT.available()) {

    char cmd = BT.read();

    Serial.print("Command received: ");
    Serial.println(cmd);

    // LED ON
    if (cmd == 'L' || cmd == 'l') {

      digitalWrite(ledPin, HIGH);

      Serial.println("LED turned ON");
      BT.println("LED turned ON");
    }

    // LED OFF
    else if (cmd == 'O' || cmd == 'o') {
```

```
  digitalWrite(ledPin, LOW);

  Serial.println("LED turned OFF");
  BT.println("LED turned OFF");
 }

// Read Temperature
else if (cmd == 'T' || cmd == 't') {

  int adc = analogRead(tempPin);

  float voltage = adc * (5.0 / 1023.0);
  float tempC = voltage * 100.0;

  Serial.print("Temperature: ");
  Serial.print(tempC, 2);
  Serial.println(" C");

  BT.print("Temperature: ");
  BT.print(tempC, 2);
  BT.println(" C");

  // Temperature Alert
  if (tempC > alertTemp) {

    Serial.println("ALERT: High Temperature!");
    BT.println("ALERT: High Temperature!");
   }

  else {

    Serial.println("Temperature is Normal");
    BT.println("Temperature is Normal");
   }
 }

// System Status
else if (cmd == 'S' || cmd == 's') {

  Serial.print("LED Status: ");

  if (digitalRead(ledPin) == HIGH) {
    Serial.println("ON");
   }

  else {
    Serial.println("OFF");
   }
```

```
      Serial.println("System Ready!");
      BT.println("System Ready!");
    }

    // Invalid Command
    else {

      Serial.println("Invalid command!");
      BT.println("Invalid command!");
    }
  }
}
```

Output –

